



777, 785, 789 and 793 OHT Disc Frictions Protection during Pre-washing

Maintenance and
Repair

Application

Component Life
Management

Component
Renewal (CRC)

MARC
Management

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1.0 Introduction

A common practice in the repair process of a component is to reuse the component part where possible.

Friction discs, which are part of the brake packs, can be reused under the conditions that are referenced in the guide of reusability SEBF8095-06 Service Brakes for Off-Highway Trucks and Tractors. The guide notes that friction discs that are exposed to water can be damaged and must be replaced, so it is necessary to protect them from water in the pre-wash area.

This Best Practice suggests the use of a brake group protection tool for trucks during the pre-wash process.



Figure 1. Brake group exposed.

2.0 Best Practice Description

Friction discs are made of different materials depending on the application and equipment (see Figure 2 and Table 1).

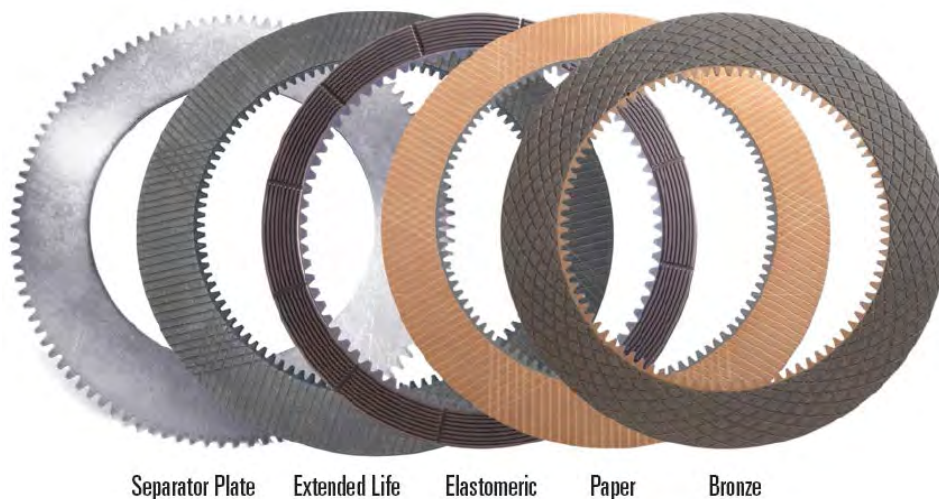


Figure 2. Caterpillar friction disc materials

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The paper disc is made of a chain cellulose (paper) and, for that reason, exposure to water causes the disc to expand thus damaging its adhesive layer. In addition, the friction material may easily come off during handling and/or installation.

Caterpillar prohibits the use of water, steam, water based solutions, freon, thinner, or ketones to clean the friction discs. It is recommended to clean with a chlorinated solvent cleaner, diesel fuel, kerosene or petroleum based distillates.

Friction Material Uses

	MACHINES	APPLICATIONS
Paper	Off-highway trucks Wheel loaders Track-type tractors	Brakes Transmission clutches Steering clutches
Bronze	Track-type tractors	Brakes
Elastomeric	Off-highway trucks Wheel loaders	Transmission clutches
Extended Life	Off-highway trucks Wheel loaders	Brakes
Separator Plate	All machines	All applications

Table 1. Uses of Friction Material

Given the damage to the paper disc that is caused by exposure to water, we developed a cover that protects the component during the prewash (Figure 3).

The protective cover includes a circular sheet, steel ring, duo-cone seal and O-ring. The plate is jointed to the circular sheet by welding points and sealed using epoxy (Figures 4 and 5). Between the seal and plate, it is also epoxied. The O-ring ensures the cover is sealed (See Figure 7).

For a quick setup the cover has a couple of grab handles.

There are three sizes currently in operation (depending on the brake package to be washed) but the design can be adapted to other components using the duo-cone seal respective (Figure 6).

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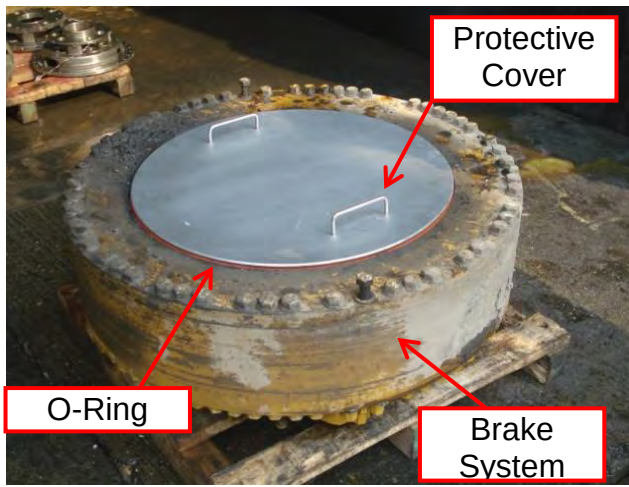


Figure 3. Protective Cover Installed.



Figure 4. Reinforced duo-cone seal.

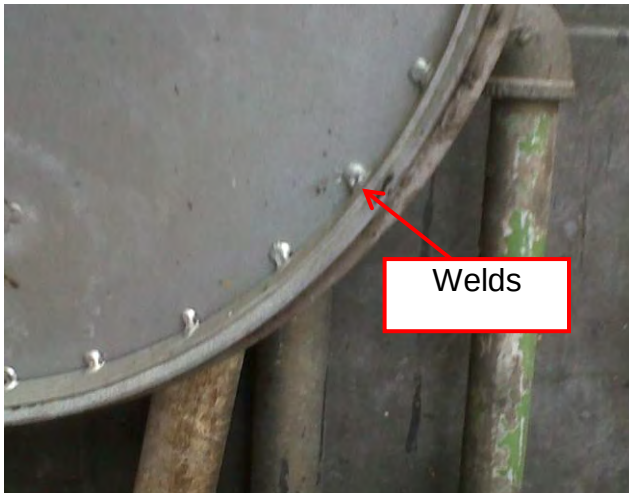


Figure 5. Duo-cone sealed lid by welding.



Figure 6. Different size protective covers.

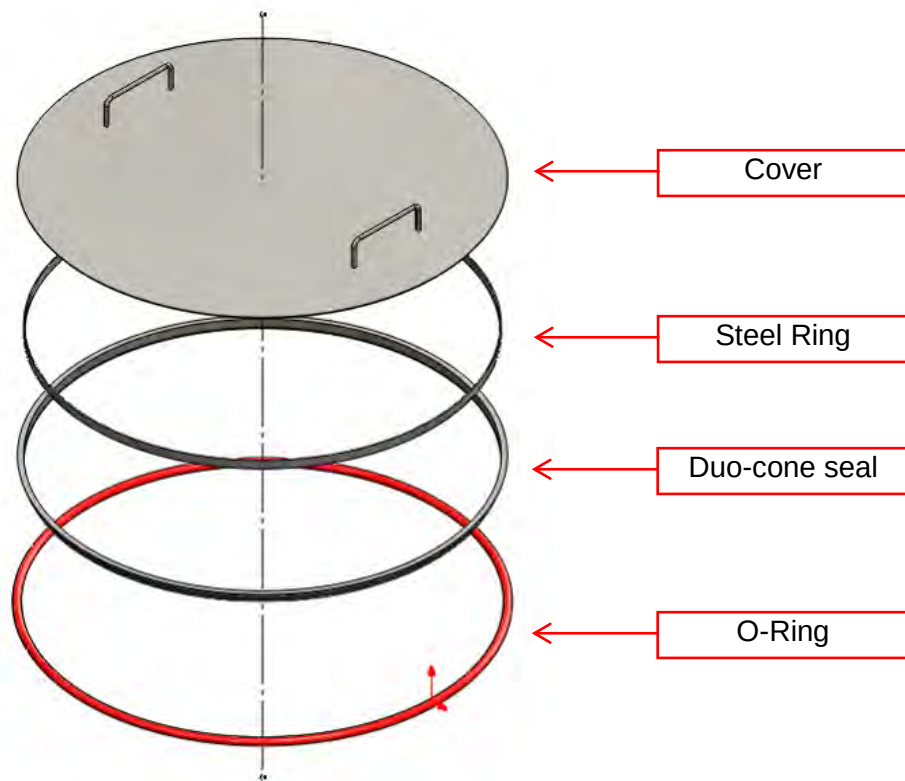


Figure 7. Assembly of protective cover.

3.0 Implementation Steps

1. Identify the model of the machine to determine the brake pack.
2. Select the correct protective cover and O-ring according to Table 2.
3. Place the protective cover using the handles.
4. Adjust for a tight fit to prevent water from entering the brake housing.

Note: The tool should be washed before and after use.

4.0 Benefits

- The discs are fully protected from water.
- Increases the reusability rate of the paper discs.
- Protect the other internal components of the brake pack.

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5.0 Resources Required

Construction of the protective cover:

- Duo-Cone seal, dimensions: Outside Diameter (see figure 7 and table 2).
- Circular steel cover, dimensions: Outside Diameter x $\frac{1}{16}$ inch.
- Steel Ring, dimensions: $\frac{5}{8}$ x $\frac{1}{16}$ inches, inner circumference equal to seal.
- Steel Rod (grab handles), dimensions: $\frac{1}{4}$ x 7 inches.
- O-ring. Selected according to duo-cone seal.
- Epoxy.
- Equipment and personnel required to construct protective cover.

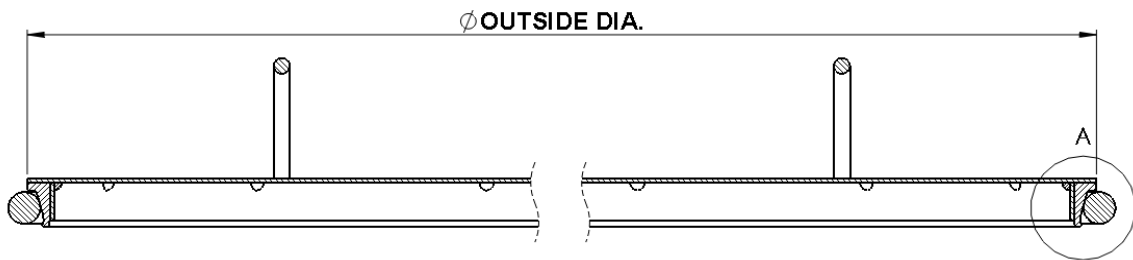


Figure 7. Profile view of protective cover.

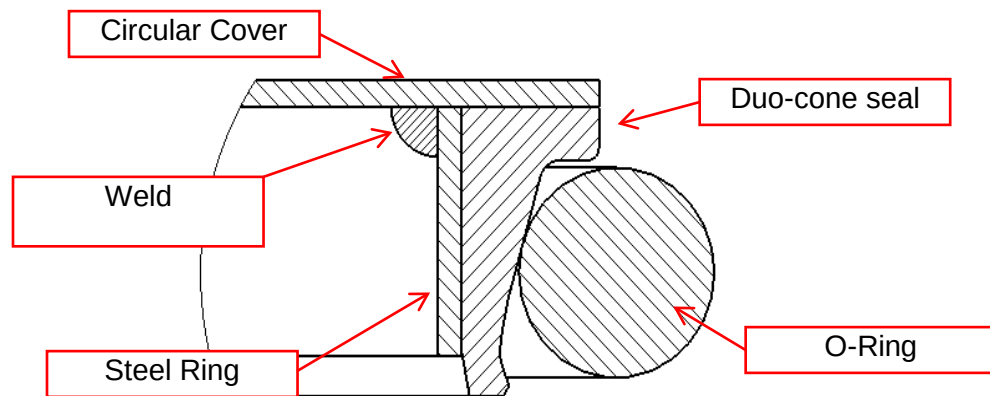


Figure 8. Detail A.

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MODEL	POSITION	BRAKE SERVICE & PARKING GP	SEAL DUO CONE GP	O-RING	PROTECTOR COVER	OUTSIDE DIAMETER (mm)
777D	FRONT	8X-3810 247-0066	297-9546	297-9545	C1-Ø21	533.40
	REAR	119-9014 247-0064				
777F	FRONT	243-5550 261-2357	297-9546	297-9545	C1-Ø21	533.40
	REAR	243-5545 261-2355 261-5545 308-3208 355-9680	365-4920	317-6440	C2-Ø22	567.94
777G	FRONT	379-1089 383-1183	365-4920	317-6440	C2-Ø22	567.94
	REAR	379-1088 383-1182				
785C, 785D	FRONT	5T-3280 240-4052 250-4052 257-6302 311-0517	297-9546	297-9545	C1-Ø21	533.40
	REAR	6G-9385 240-4051 245-6467 257-6300 311-0516	314-4130	314-4129	C3-Ø27	700.00
789B, 789C, 789D	FRONT	5T-5070 106-9885 250-4049 257-6301	297-9546	297-9545	C1-Ø21	533.40
	REAR	5T-4489 164-6607 198-1911 250-4048 257-6298	314-4130	314-4129	C3-Ø27	700.00
793B, 793C, 793D, 793F	FRONT	8X-0222 219-0325 245-6466 250-3938 252-8709 334-5020	314-4130	314-4129	C3-Ø27	700.00
	REAR	8X-0241 197-9554 250-3940 334-5019				
793D XLP, 793F XLP	FRONT	188-5020	378-0592	314-4135	C4-Ø32	806.70
	REAR	188-5019				

Table 2. Part numbers and dimensions

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6.0 Supporting Attachments / References

- SIS document: SEBF8095-06 (Service Brakes for Off-Highway Trucks and Tractors)
- Cat Friction Material PEHJ0031-01
- SIS document: SEHS8484-17 (Tool and Specification Chart for Duo-Cone Seal)

7.0 Related Best Practices

8.0 Acknowledgements

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